Odd End Semester Supplementary Examination -2024

M.TECH (Geo-Informatics)
Semester:I

Name of the Course: M.Tech **Semester : I**
Name of the Paper :Probability and Statistics For Engineering **Paper Code : MGI 103**
Time: 3Hour **Maximum Marks: 100**

NOTE:

(i) All questions are **compulsory.**
(ii) Answer any two sub questions among a,b & c in each main question.
(iii) Total marks in each main question are **twenty.**
(iv) Each question carries 10 marks.

Q1 **(20 marks)**

(a) Calculate median from the following frequency distribution.

Class intervals:	0-10	10-30	30-60	60-80	80-90	90-100
No. of students	5	16	30	12	6	1

(b) An urn I contains 3 white and 4 red balls and an urn II contains 5 white and 6 red balls. One ball is drawn at random from one of them urns and is found to be white. Find the probability that it was from urn I. **CO1**

(c) Define Range. Calculate Quartile deviation and Mean deviation from the following data.

Age:	10	15	20	30	40	50
Frequency:	8	12	15	10	3	2

Q2 **(20 marks)**

(a) The diameter of an electric cable is assumed to be continuous random variate with probability density function:
$f(x) = 6x(1-x) \quad 0 \le x \le 1$ verify that above is p.d. f. also finds mean and variance.
.

CO2

(b) Define probability density function and probability mass function. A discrete random variable X has the probability function given below:
Values of

Values of x:	0	1	2	3	4	5	6	7
P(x)	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2+k$

(i) Find k

(ii) Find P(X< 6), P(0< X<4) and P (X> 6)

.

(c) Ten coins are tossed simultaneously Find the probability of getting at least 7 heads.

Q3 **(20 marks)**

(a) Write a short note on Estimation Theory. **CO3**

(b) In experience on pea- breeding Mendal obtained the following frequencies of seeds:

Red & Yellow	Wrinkled & yellow	Round & Green	Wrinkled & green	Total
315	101	108	32	556

Theory predicts that the frequencies should be in proportion 9: 3: 3:1. Examine the correspondence between theory and experiment. Give that the value of χ^2 for 3 degree of freedom at 5% level of significance is 7.815. **CO5**

(c) Define Interpolation. Using Newton's forward difference interpolation formula, find the form of the function from the following data: **CO3**

0:	3	2	1	-1
f(x)	3	12	15	-21

Q4 **(20 marks)**

(a) 4 coins were tossed 160 times and the following results were obtained: **CO4**

No. of heads	0	1	2	3	4
Observed frequency	17	52	54	31	6

Under the assumption that coins are unbiased, find the expected frequencies of getting 0,1,2,3 or 4 heads and test the goodness of fit.

(b) Write a short not on chi-square test of goodness of fit and chi-square test for independence of attributes.

(c) The following data related to the heights (in cms) of two different varieties of wheat plants.

Variety 1	92	100	85	94	95	98	96	100
Variety 2	75	70	80	85	78	80	-	-

Test the hypothesis that mean heights of plants of both varieties are the same.

Q5 **(20 marks)**

(a) Discuss the meaning of correlation. Find correlation by Kearl's Pearson coefficient method by using the given data. **CO5**

X	10	14	18	22	26	30
Y	18	12	24	6	30	36

(b) What is principle of least squares? Fit a second degree parabola from the given data.

x:	1	2	3	4
y:	6	11	18	27

(c) From the following data obtain regression of y on x :

X	1.53	1.78	2.60	2.95	3.42
Y	33.50	36.30	40.00	45.80	53.50